
Software Requirements Specification

for

CASTLE

Version 1.0 approved

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Table of Contents

Table of Contents.....	2
Revision History.....	3
1. Introduction.....	1
1.1 Purpose.....	1
1.2 Document Conventions.....	1
1.3 Intended Audience and Reading Suggestions.....	1
1.5 References.....	2
2. Overall Description.....	2
2.1 Product Perspective.....	2
2.2 Product Function.....	3
2.3 User Classes and Characteristics.....	3
2.4 Operating Environment.....	3
2.4.1 Hardware Platform.....	3
2.4.2 Operating Systems and Versions.....	4
2.4.3 Software Components.....	4
2.5 Design Implementation Constraints.....	4
2.6 User Documentation.....	4
2.7 Assumptions and Dependencies.....	5
2.7.1 Assumptions.....	5
2.7.2 Dependencies.....	5
3. External Interface Requirements.....	5
3.1 User Interfaces.....	5
3.2 Hardware Interfaces.....	5
3.3 Software Interfaces.....	6
3.4 Communication Interfaces.....	6
4. System Features.....	6
4.1 Topology Detecting and Handling of a Sand-Based Surface.....	6
4.1.1 Description and Priority.....	6
4.1.2 Stimulus Response Sequences.....	6
4.1.3 Functional Requirements.....	7
4.2 Responsive Projection onto Sand-Based Surface.....	7
4.2.1 Description and Priority.....	7
4.2.2 Stimulus Response Sequences.....	7
4.2.3 Functional Requirements.....	8
4.3 Secure and Locking Body.....	8
4.3.1 Description and Priority.....	8
4.3.2 Stimulus Response Sequences.....	8
4.3.3 Functional Requirements.....	9

4.4 Mobility of System.....	9
4.4.1 Description and Priority.....	9
4.4.2 Stimulus Response Sequences.....	9
4.4.3 Functional Requirements.....	10
4.5 Lesson Plan Library.....	10
4.5.1 Description and Priority.....	10
4.5.2 Stimulus Response Sequences.....	10
4.5.3 Functional Requirements.....	11
5. Other Nonfunctional Requirements.....	11
5.1 Performance Requirements.....	11
5.1.1 Performance Requirement List.....	11
5.2 Safety & Security Requirements.....	12
5.2.1 Safety & Security Requirements List.....	12
5.3 Software Quality Attributes.....	12
5.3.1 Software Quality Attribute List.....	12
5.4 Business Rules.....	13
5.4.1 Business Rule List.....	13
6. Other Requirements.....	13
Appendix A: Glossary.....	13
Appendix C: To Be Determined List.....	15

Revision History

Name	Date	Reason for Changes	Version

1. Introduction

1.1 Purpose

The purpose of this document is to provide a detailed description for CASTLE, which includes how it functions, the main objective of the project, and constraints the project has. CASTLE uses a variety of hardware such as a depth camera and sensors to track hand movement in the sand and how the sand is moved. It also uses a projector to project lesson scenarios onto the sand. The lesson scenarios are provided from an external source. C++ is the main language used for the software side of CASTLE, which we use to implement and simulate the project.

1.2 Document Conventions

This document uses an IEEE template provided by our professor Dr. Sakka of CS/EE/CPE-480. This document uses Arial font and the following typographical conventions:

- 20pt font is used for Section headers
- 16pt is for headers in the section (example 1.1)
- 14pt font is for subheaders within headers (example 2.4.1)
- 11pt font for the main body of each section
- Priority levels are defined as **HIGH**, *Medium*, and Low
- Requirements are labeled with different identifiers (example FR-1)

1.3 Intended Audience and Reading Suggestions

This document is intended for the developers of the project, and WVU Tech professors such as the professors for CS/EE/CPE-480 along with advisors. The developers of the project should use this document as a guide on how CASTLE works such as how the hardware and software are implemented along with knowing how to test it properly. WVU Tech instructors for CS/EE/CPE-480 should be expected to review this document to understand how CASTLE functions with other professors reviewing this document for recommendations for any changes to make.

To fully understand this document it is recommended to start at the Introduction section followed by the Overall Description section as this will allow the reader to have a stronger understanding of the overall project and its goals. While reading any section of this document there may be terms that may be hard to interpret, it is recommended to go to Appendix A which is the glossary for this SRS in which a hard to interpret term is defined.

1.4 Product Scope

This document provides a detailed plan to develop CASTLE as a learning tool to assist neurodivergent students in schools, in which it uses a sandbox approach for tactile learning. It will use a projector that projects lessons onto the sand with depth cameras and sensors being used to track things like hand movement, and user-altered sand topology.. This could include things such as spelling a word, or solving basic arithmetic problems. For the lessons it will use programs made specifically for CASTLE and would be provided externally. For the purposes of this project however, only a handful of lessons will be provided for the initial prototype. Since lessons are provided externally, and the aim for CASTLE is to be modular, each lesson is treated as an independent portion of the internal software.

This project aligns with a growing trend to implement more technology based approaches to assist in the learning of special education students rather than using traditional paper and pencil techniques [1]. It also aligns on how most neurodivergent students learn, in which they learn through feeling the environment around them hence the tactile approach of using a sandbox approach [2].

1.5 References

[1] K. J. V. Matias, C. A. Villanoza, R. T. Estabillo, and M. W. Lattao, "Mapping the Intellectual Landscape of Inclusive Education for Neurodivergent Learners," *Journal of Learning and Development Studies*, vol. 6, no. 6, pp. 8-20, Apr. 2026.

[2] S. Tancredi, "Sensory-adaptive embodied design: Integrating movement for sensory regulation and conceptual learning," *Multimodality & Society*, University of Wisconsin–Madison, 2026.

2. Overall Description

2.1 Product Perspective

CASTLE is a new, self contained learning system that is designed to serve as an active, physical learning environment for neurodivergent students. The system combines a physical sandbox, depth camera, projector, computer hardware, speakers, and software into an interactive system.

The software serves as the primary interface between the depth camera and feedback provided to the user on the projector and speakers. The depth camera will capture information from the surface of the sand as well as hand gestures, and provide that information to the computer hardware. The software will process this information and provide feedback to the user through the projector by displaying images on the sand and auditory sounds through speakers as illustrated in Figure 1. This system is intended to function in an education environment and serve as an aid to learning along existing curriculum.

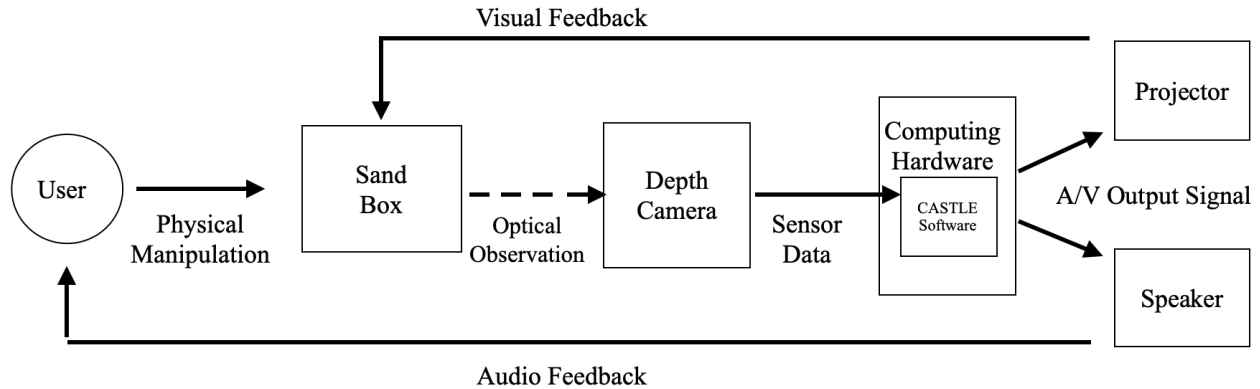


Figure 1: System Architecture and Data Flow of CASTLE

2.2 Product Function

- User Interaction
- Data Processing
- Sand Surface Detection
- Visual Feedback
- Audio Feedback
- System Configuration

2.3 User Classes and Characteristics

1. **Primary User (Students):** This class directly interacts with the physical learning environment by manipulating the sand which triggers audio and visual feedback. This system is designed for neurodivergent students for tactile learning therefore, this class requires no prior technical expertise or software training. Their interaction with the sandbox is physical, sensory, and intuitive.
2. **Secondary Users (Facilitators/Educators):** These users will act as facilitators for the system. They will interface with the system configuration to boot up the computer hardware, launch the application, and select lesson modules. This class requires a basic to intermediate level of computer literacy. They require a user friendly interface that allows them to manage the system without the need to interact with the raw data or code.
3. **Developers:** These users require advanced computer literacy. They will develop lesson plans and upload them to the existing library.

2.4 Operating Environment

2.4.1 Hardware Platform

- **Computer:** The software shall operate on a computer capable of processing a depth camera, projector, and other peripheral devices.

- **Depth Camera:** The system will interface with a depth sensing camera to gather information from the sand surface and user interactions.
- **Projector:** The system will interface with a projector to provide visual feedback and educational content displayed on the sand.
- **Speakers:** The system will interface with speakers to provide audio feedback and educational content to users.
- **Sandbox:** The system will operate within the physical dimensions and configuration of the sandbox structure.

2.4.2 Operating Systems and Versions

The software will operate on an operating system supported by the selected computer program. The current proposed system is Linux.

2.4.3 Software Components

The software will include the appropriate interfaces to interact with the depth camera, projector, and other peripheral devices. Additional software libraries for depth sensing, image processing, visualization, and audio output will be identified as software architecture is finalized.

2.5 Design Implementation Constraints

- **Hardware Compatibility:** The software shall be compatible with the selected depth camera, projector, speakers, sandbox, computer, and other peripheral devices. The software shall account for the physical dimensions of the sandbox as well as the positioning of the depth camera and projector. Calibration shall be considered to ensure that the detected features and displayed content appropriately align with the physical dimensions of the sandbox.
- **Software, Technology, and Performance Constraints:** The software shall operate within the capabilities and requirements of the operating system. The system shall process depth camera information and provide audio and visual feedback with sufficient response time to support interactive experience. Specific programming languages will be selected based on compatibility with the system hardware and functional requirements.
- **Budget and Resources Constraints:** The implementation will remain within the available budget and resources available to the senior design team.
- **Safety Constraints:** The system shall be designed to operate safely.

2.6 User Documentation

An in-App user guide, tutorial videos, and written manual will be provided explaining how to power up the system, choose learning modules, calibrate the depth camera, and perform routine maintenance.

2.7 Assumptions and Dependencies

2.7.1 Assumptions

1. The depth camera will be sufficient for detecting changes in the sand and user interaction.
2. The projector will display sufficient resolution and brightness.
3. The selected computer will have sufficient capabilities to operate the depth camera, projector, speakers, and software simultaneously.
4. The sandbox structure will be constructed to the appropriate dimensions and stability.
5. Users will interact with the system under normal operating conditions.

2.7.2 Dependencies

1. The system depends on the selected operating system to coordinate between the depth camera, speakers, and projector.
2. The system depends on a depth camera to capture and interpret surface and gesture data.
3. The system depends on a projector for displaying visual content.
4. The system depends on successful calibration to function properly.
5. The system depends on access to a continuous source of power.

3. External Interface Requirements

3.1 User Interfaces

For the user interface we will have a basic layout. The system will automatically launch to the calibration screen for the projector and camera. After calibration it will go to the home page gui this will consist of the lesson, help, and recalibrate buttons. Clicking the lessons button will launch to the lesson selection screen, this will show all currently available lessons as soon as one is clicked it will start on the sand box. For the help button it will open a menu consisting of tips and directions to the help documentation. The recalibrate selection will be password protected in case a child would get in contact with the screen.

3.2 Hardware Interfaces

In this project we use three main hardware interfaces: a depth camera, projector, and computer hardware. The depth camera is used to capture information about the grooves in sand made by interaction. The computer hardware is used to run the software and process the data, as well it enables communication between the depth camera and projector. The projector is used to display the visual information from the depth sensor.

3.3 Software Interfaces

The software will be written in QML and C++. QML will be used to create the graphical user interface, and C++ will be used for logical computation and communication between the hardware and software. OpenCV, an open source computer vision library, will be used for the camera calibration and image processing for the projector. The calibration data will make it possible for the software to map points from the depth camera to the corresponding positions on the sand surface for the projector to place the appropriate imagery.

3.4 Communication Interfaces

CASTLE will require a network connection for teachers to upload custom lessons to a server, as well as download lessons uploaded by other teachers. The software hosted on the computer hardware will connect with the server to transfer the files. Teachers will be able to download lessons from the server for future usage. The depth camera and projector will automatically be able to understand any new lesson run on the system.

4. System Features

4.1 Topology Detecting and Handling of a Sand-Based Surface

4.1.1 Description and Priority

The system will detect the three dimensional topology of the sand using depth sensing hardware. This data will be processed to create a digital representation of the surface. Using this representation the system will update accordingly, maintaining cohesion between the projected visuals and the physical surface.

HIGH PRIORITY: The functional detection and handling of the surface of the sand is the core of the system. This model is required for both interaction with the system as well as the digital output of the system. Should this feature fail, the system would be severely limited in its capability.

4.1.2 Stimulus Response Sequences

Stimulus: A user makes a physical change to the surface of the sand.

Response: The system will detect the physical change to the elevation of the sand and update the internal representation of the terrain. The visual response of the system will adapt to the updated state and refresh the projected visual.

4.1.3 Functional Requirements

- FR-1: The system will detect the physical state of the sand within the confines of the sandbox area
- FR-2: The system will measure the changes in the elevation of the sand using depth sensing hardware
- FR-3: The system will store the measured changes in a digital representation of the sandbox's topology
- FR-4: The system will update the digital representation of the topology when changes are made to the surface of the sandbox.
- FR-5: The system will provide the digital representation of the topology to the visual projected subsystem
- FR-6: The system will be able to make a distinction between the defined area of the sandbox surface and the bounds that are outside of the designated sandbox surface.
- FR-7: The system will be able to maintain constant updates to the terrain representation while within continuous system use.

4.2 Responsive Projection onto Sand-Based Surface

4.2.1 Description and Priority

The system will be able to handle a given digital representation of the three dimensional surface of the sand and display a projected visualization accordingly. The projected image will correspond to the detected terrain of the system, accounting for differences in the surface elevation of the sandbox and overlaying the image in the correct location. The system will also be able to display the system configurations and settings within the sand, projecting menus and configurations, as well as every visual needed for system use and management.

PRIORITY HIGH: The visual feedback of the system, the projected visuals are the foundation of user feedback from the system. Providing an experience of the system. Failure of this system will significantly impact the user experience and operation of the system.

4.2.2 Stimulus Response Sequences

Stimulus: The internal terrain representation is updated after a change is made to the sand surface

Response: The projection subsurface will generate an updated visualization to correspond with the new terrain state and project the updated image to the appropriate location of the sandbox.

Stimulus: a user selects or activates the system menu interface

Response: The projection subsystem will display the corresponding menu, controls, or configuration information to the appropriate location of the sandbox.

4.2.3 Functional Requirements

- FR-8: The system will be able to receive the current digital representation of the sandbox topology from the topology subsystem.
- FR-9: The system will generate a projected visualization that corresponds to the changed elevation of the sandbox
- FR-10: The system will align the projected visuals to their corresponding physical location within the sandbox
- FR-11: The system will be able to change the projected visuals with each update of the terrain representation.
- FR-12: The system will be able to display system menus, configurations, and other required user interface elements.
- FR-13: The system will be able to maintain continuous visual output during normal operations.

4.3 Secure and Locking Body

4.3.1 Description and Priority

The system will contain all electrical components, physical connections, system hardware, and computational components within the confines of a secure and enclosed body. The enclosure will prevent unauthorized users from the direct access of internal components during normal operations. The enclosure will also allow authorized access to the system for maintenance, repair, and configuration. All removable portions of the enclosure will incorporate locking mechanisms to prevent unauthorized or unintended access. The enclosure will also provide physical protection for the internal wiring, computer hardware, power distribution components and other sensitive equipment.

PRIORITY HIGH: The enclosed body and locking features of it provide safety not for just the components themselves but also to the users of the system. Because the system is intended for use in a classroom environment, exploded electrical and computation components present safety concerns as well as risk of damaging the system. Failure of the enclosure or locking mechanism could compromise the safety, reliability, or continued use of the system.

4.3.2 Stimulus Response Sequences

Stimulus: A user attempts to access a secured portion of the enclosure without unlocking the system.

Response: The enclosure will remain physically secure, preventing access to internal components

Stimulus: An authorized user unlocks a secured access panel or lid for maintenance

Response: The system will allow access to the designated internal components while maintaining its structural integrity.

Stimulus: A secured access point is returned to its normal operating position.

Response: The locking mechanism will allow the access point to be secured before normal system operations resume.

4.3.3 Functional Requirements

- FR-14: The system will enclose internal electrical, computational, and mechanical components within the body of the system.
- FR-15: The system will prevent direct access to internal components by a normal user during operation
- FR-16: The system will include a locking mechanism on all lids, access panels or maintenance openings.
- FR-17: The system will allow access to any authorized person for maintenance and repair.
- FR-18: The system will protect the internal wiring and electrical connections from accidental physical contact during normal use.
- FR-19: The system will maintain structural stability when access panels, and or lids, are in the open or closed position
- FR-20: The system will prevent access panels from opening during the normal use of the system.

4.4 Mobility of System

4.4.1 Description and Priority

The system will be designed to allow transportation of the system between locations without the need of major disassembly. The complete system will incorporate mobility features that allow authorized users to reposition or transport the system with minimal risk of damage to the enclosed or internal components, whilst maintaining rigidity during normal operation.

PRIORITY MEDIUM: Mobility of the system is an important part of classroom deployment, that impacts the user's ability to maintain, store, and relocate the system. However the system would still be able to perform primary functions while stationary. Failure of the mobility of the system would reduce convenience and flexibility but would not hinder the system's core operations

4.4.2 Stimulus Response Sequences

Stimulus: An authorized user prepares the system for relocation.

Response: The system will allow its mobility components to be placed into a state that permits controlled movement of the complete system.

Stimulus: An authorized user applies force to move the system across an indoor surface.

Response: The system will move in a controlled manner while maintaining system integrity.

Stimulus: The system is placed in its intended operating location.

Response: The system will remain stable and stationary under normal operations.

4.4.3 Functional Requirements

- FR-21: The system will include mobility components that allow it to be moved easily between appropriate indoor areas.
- FR-22: The system will allow relocation of the entire system without the need of major disassembly.
- FR-23: The system will not require alteration of electrical or computational components to be transported.
- FR-24: The mobility system will be able to support the entire weight of the operational system.
- FR-25: Movement of the system will not require access to normally secured locations of the system.
- FR-26: Internal components will not be compromised or damaged by the movement of the system.

4.5 Lesson Plan Library

4.5.1 Description and Priority

The system will include access to a library of downloadable educational content, referred to as “Lessons”. The system will come included with a set of lessons by default, while additional lessons may be made available through external open-source repositories. The system will provide users a method to browse and download available lessons.

PRIORITY MEDIUM: Access to downloadable lessons increases the educational applications of the system. Capability of the system to allow new content without the need for a system update can greatly improve user experience and product capability. Failure to implement will leave our system static, but the core features of the system will remain intact.

4.5.2 Stimulus Response Sequences

Stimulus: An authorized user attempts to access the library via the menus

Response: The system will allow the authorized user to view and download lesson plans

Stimulus: An unauthorized user attempts to access the library via the menus

Response: The system will notify the user that they do not have the proper permissions to access the library and deny access.

Stimulus: An authorized user selects the download option on a valid lesson plan.

Response: The system downloads the available content and stores it in the correct directory.

Stimulus: An authorized user selects the delete option on a valid lesson plan.

Response: The system locates the corresponding lesson plan in the system's directory and removes only the file specified to delete.

4.5.3 Functional Requirements

- FR-27: The system will provide an interface allowing authorized users to access the lesson library.
- FR-28: The system will display a list of available lesson plans within the lesson library.
- FR-29: The system will allow authorized users to view information associated with available lesson plans before downloading.
- FR-30: The system will allow authorized users to download selected lesson plans from available repositories.
- FR-31: The system will store downloaded lesson plans within the designated system directory.
- FR-32: The system will verify that downloaded lesson files are successfully stored before making them available for use.
- FR-33: The system will allow authorized users to remove previously downloaded lesson plans from the system.
- FR-34: The system will only remove the lesson plan file specifically selected by the authorized user.
- FR-35: The system will prevent unauthorized users from accessing lesson library management functions.
- FR-36: The system will notify users when they attempt to access restricted lesson library functions without proper authorization.
- FR-37: The system will support the addition of new lesson content without requiring a complete system software update.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

CASTLE is intended to provide an interactive learning environment in which changes made within the sandbox produce some kind of response whether visual or auditory. The system needs to maintain consistent performance while handling user input and system output.

5.1.1 Performance Requirement List

- PR-1: The system should be able to accurately process sand depth throughout the entire lifespan of CASTLE's equipment being turned on.
- PR-2: The system should feel responsive, and undelayed.

- PR-3: The operation of each component, in both software and hardware, should work in tandem at all times. (No requirement of suspending components).
- PR-4: The system should have accurate measurements of the 'current state' of the sandbox at all times.

5.2 Safety & Security Requirements

CASTLE will operate within classroom environments and will have direct physical interaction with students. The system has to maintain complete safe operating conditions during normal use, and students will have limited access to components or system functionality that could create unsafe conditions.

5.2.1 Safety & Security Requirements List

- SSR-1: The system will have zero accessibility to direct contact with powered electrical components during normal operation.
- SSR-2: The physical system should maintain stability throughout any general interaction. (Not tippable).
- SSR-3: The physical space holding 'play-sand' should be completely isolated, and should not interrupt electrical or computational components within the system.
- SSR-4 The system should have both a safe start-up, and shut-down no matter its current state.

5.3 Software Quality Attributes

CASTLE's software will be designed to operate reliably within a classroom environment. The foundational software design will be modular, allowing new components of the system to be developed, maintained and modified independently.

5.3.1 Software Quality Attribute List

- SQA-1 – Reliability: Software should maintain a stable operation during normal use without loss of current system state.
- SQA-2 – Usability: Software user-interface should allow educators with basic computer literacy to operate the primary functions of CASTLE without needing a deeper understanding of the system.
- SQA-3 – Maintainability: The software will be organized such that subsystems can be modified, tested, and maintained without needing excessive modification of unrelated systems.
- SQA-4 – Modularity: Input and output software functions should be treated separately with defined interactions between them to ensure modification capabilities, and creation of new 'lesson-plans'.
- SQA-5 – Extensibility: The addition of new lesson modules will not require modification of core functionality of CASTLE, like keeping the current sandbox state stored.

5.4 Business Rules

Operation within an educational environment will mean students and educators will have different roles to feel within CASTLE's system. Students will interact with the physical sandbox, and lesson activities. Educators and authorized individuals will be responsible for functions like lesson selections, calibration, and maintenance.

5.4.1 Business Rule List

- BR-1: Students can interact with the physical sandbox environment, but will not have access to system configuration or calibration.
- BR-2: Educators may manage lesson content.
- BR-3: Physical maintenance of CASTLE will only be performed by authorized individuals while the system is not currently being used by students.

6. Other Requirements

Appendix A: Glossary

Authorized User: A person permitted to access secured portions of the system for maintenance, repair, configuration, or transportation.

CASTLE: Class Room Adaptive Tactile Learning Environment.

Calibration: The process of comparing a measuring device or instrument against a standard reference to check its accuracy and find any errors.

computer hardware: The physical, tangible components and equipment that make up a computer system and facilitate the execution of software.

Depth Camera: A specialized imaging device that measures the distance from the camera to objects in a scene.

Designated Sandbox Surface: The defined physical region of sand that the system recognizes and processes.

Digital Terrain Representation: The internal software representation of the detected sand surface, including its elevation or depth data.

IEEE: Institute of Electrical and Electronics Engineers.

Mobility Components — Mechanical components that allow the complete system to be moved or repositioned, such as casters or wheels.

Neurodivergent: Describing individuals whose brain function or cognitive processing differs from neurotypical norms (e.g., individuals with Autism Spectrum Disorder, ADHD, or sensory processing differences).

Normal Operation: The intended state of the system while it is being used without maintenance, repair, or transportation activities.

Normal User : A person interacting with the system during ordinary operation who is not expected to access internal hardware.

OpenCV (Open Source Computer Vision Library): An open source software library used for real-time computer vision, image processing, and camera calibration.

Projection Alignment: The process of ensuring that generated visuals correspond to the correct physical locations within the sandbox.

QML (Qt Modeling Language): A user interface markup language used to design responsive UI components in C++ applications frameworks.

Responsive Projection: Projected visual output that changes in response to changes in the detected sand surface or user interaction.

Stimulus-Response Sequence: A defined interaction loop where an environmental or user action (stimulus) triggers a corresponding system calculation and physical/visual feedback (response).

Subsystem: A logical or functional portion of the overall CASTLE system responsible for a specific set of operations.

System Enclosure: The physical body surrounding and protecting the system's electrical, computational, and mechanical components.

Tactile Learning: A hands-on instructional methodology where individuals learn through physical interaction, touch, and manipulation of physical media.

Topology: The three-dimensional surface shape, elevation changes, and structural contours of the sand inside the sandbox area.

Appendix C: To Be Determined List

➤ TBD-1: Final Computing Hardware Specifications

Description:

The final specifications for the computing hardware required to operate CASTLE are not yet finalized. The hardware must be selected such that it can support real-time processing of the depth-camera, projector, lesson modules, and system software.

Reason:

The hardware requirements will depend on the finalization and optimization of software.

➤ TBD-2: Depth Camera Calibration Software

Description:

The calibration procedure between the depth camera, projector, and sandbox surface has not yet been finalized

Reason:

Calibration requirements will depend on physical placement of camera, projector, and sandbox dimensions which have not yet been assembled.

➤ TBD-3: Lesson Plan File Format and Structure

Description:

The final format that is used to store, organize, load and execute lesson plans has not yet been determined.

Reason:

The lesson plan architecture depends on finalized software of how the lessons interact and communicate with the core CASTLE system.

➤ TBD-4: Lesson Library Hosting and Distribution Method

Description:

The final method for storing and distributing external lesson content has not yet been determined.

Reason:

The system plans to support external lesson sources, but the exact repository, server structure, and communication method will need to be optimized to best fit the lesson plan file format and structure which have still not yet been defined.

➤ TBD-5: Exact Performance Requirements

Description:

The measurable performance requirements for system response time, depth processing rate, and projection update frequency have yet to be defined.

Reason:

Performance requirements are dependent on hardware testing and software optimization which have not yet been integrated.